

12
CHAPTER

DNS (Domain Name System)

BRSVJM

LEARNING OBJECTIVES

- To understand the need of Domain Name System for proper functioning of Internet
- To know the importance of IP addresses
- To know the parts of URL and its types
- To know the components of Domain name system and its functions
- To know how the DNS is working

12.1 Introduction

In earlier days, websites were accessed through their IP addresses. It was difficult for an individual to remember all the IP address to access the websites. So, the domain names were created and mapped with IP addresses. Like phone book, where all the contact numbers are stored under respective names and are accessed by the contact names. Domain Name System (DNS) maintains all the directory of domain names/host names and help us to access the websites using the domain/ host names.

12.2 Overview of DNS

For the communication to takes place, the information should pass through seven layers. Application layer is one among the seven layers. There are several applications in the application layer and DNS (Domain Name System) is one among them. Internet is based on IP addresses, not domain names. But it is easy to use domain names to refer them rather than the long numbers (IP address). To enable the use of domain names in a network, the Domain Name System (DNS) is used. Domain names to IP address mapping must be consistent across the network to ensure interoperability. DNS provides the domain name to IP address mapping through Name servers. To know more about DNS working we must first know about IP address, URL and DNS components. So, let us see about each in detail. Refer Figure 12.1

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